

SUMMARY

Data Scientist and Generative AI / ML Engineer with hands-on experience in designing, developing, and deploying scalable **machine learning models**, **LLM applications**, and **AI solutions** across **cloud environments**. Strong background in **supervised and unsupervised learning**, **NLP**, **computer vision**, **deep learning**, and **MLOps**. Proficient in leveraging **Gen AI frameworks**, **transformer models**, and **vector databases** to build **RAG systems**, **chatbots**, and **LLM fine-tuning pipelines**. Experienced in **data preprocessing**, **model training**, **A/B testing**, **prompt engineering**, and **integrating AI services** with **REST APIs**. Demonstrated ability to improve model performance by over **30%**, reduce processing time **by 40%**, and increase **data pipeline efficiency by 45%**.

SKILLS

- **Languages & Frameworks:** Python, PyTorch, TensorFlow, Hugging Face Transformers, LangChain, OpenAI, Scikit-learn, FastAPI, Flask.
- **Machine Learning & Gen AI:** LLMs, RAG, Fine-tuning, Prompt Engineering, NLP, Computer Vision, Recommendation Systems, Chatbots.
- **Data & Storage:** SQL, Pandas, NumPy, Spark, Hadoop, Hive, Delta Lake, Snowflake, BigQuery
- **MLOps & Deployment:** Docker, Kubernetes, MLflow, SageMaker, Airflow, Git, CI/CD, FastAPI, REST APIs
- **Vector DBs & Tools:** FAISS, Pinecone, Weaviate, ChromaDB, Elasticsearch, Milvus
- **Cloud Platforms:** AWS (S3, EC2, Lambda, SageMaker), GCP, Azure
- **Visualization & Analysis:** Matplotlib, Seaborn, Plotly, Tableau, Power BI

EDUCATION

- Master of Science in Computer Science From Pace University, NYC

PROFESSIONAL EXPERIENCE

Meta | Data Scientist / Gen AI Engineer | Jan 2024 - Current

- Developed Retrieval-Augmented **Generation (RAG) systems** using **OpenAI**, **LangChain**, and **FAISS**, improving response accuracy by **37%**.
- Built **LLM-based** internal **chatbot** using **GPT-4**, **Pinecone**, and **LangChain**, **reducing customer service** query time **by 42%**.
- **Fine-tuned open-source models (LLaMA2, Falcon)** using domain-specific datasets and frameworks like **Hugging Face** and **NLTK**, increasing model accuracy **by 28%**.
- Deployed scalable **AI models** using **Docker**, **Kubernetes**, and **FastAPI**, reducing inference latency **by 35%**.
- Automated model **training**, **evaluation**, and **tracking** using **MLflow** and **Airflow pipelines**, **boosting workflow efficiency by 40%**.
- Engineered prompts for **LLMs** such as **GPT-4**, **Claude**, and **Gemini** to optimize **QA** and **summarization tasks**, improving output quality **by 25%**.
- Applied advanced **NLP techniques** using **NLTK**, **Transformers**, and **LangChain** for tasks like **entity recognition**, **summarization**, and **classification**.
- Performed **A/B testing** and **model performance** evaluation using **SQL** and **Power BI** to assess feature enhancements and drive adoption metrics.

Amazon | Data Scientist / AI/ML Engineer | Aug 2019 - Dec 2022

- Developed supervised **ML models** for demand **forecasting** and **personalized** recommendations using **Scikit-learn**, **LightGBM**, and **XGBoost**, improving accuracy **by 32%**.
- Built **NLP pipelines** for **text classification**, **named entity recognition**, and sentiment analysis using **BERT**, **spaCy**, and **NLTK**, increasing **F1-score by 27%**.
- Created **REST APIs** with **Flask** and **FastAPI** to serve **ML models** in real time, reducing inference latency **by 38%**.
- Implemented **computer vision** models with **TensorFlow** and **OpenCV** for **defect detection**, achieving **89% precision** on **production data**.
- Optimized **data pipelines** using **Apache Spark** and **Airflow**, increasing **data throughput by 45%** and **pipeline uptime by 33%**.
- Used **AWS SageMaker** for end-to-end **ML workflows**, reducing model deployment time **by 35%** and cloud costs **by 30%**.

- Built interactive **dashboards** with **Tableau**, **Seaborn**, and **Plotly** to **visualize model** performance and business insights.